shrubs further enrich the habitat with soil and humus. Therefore, the shrubs are replaced by trees which constitute the climax community.

**6. Forest stage** - The trees capable of growing in xerophytic condition try to invade the area which was occupied previously by shrubs. Further increasing the humus content of the soil favours the arrival of more trees and vegetation finally become mesophytes. As the trees are deeply rooted and much branched, they absorb more quantity of water and nutrients. After a long interval, a complete harmony is established among the plant communities. The climax stage remains unchanged unless some major environmental changes disturb it

Of the 6 stages of lithosere succession, stage 1 is occupied by pioneer community and the stage 6 is occupied by climax community. The stages 2 to 5 are occupied by seral communities. Seral stages occurring on the same rock surfaces.

### 7.3.6 Significance of Plant Succession

- Succession is a dynamic process. Hence an ecologist can access and study the seral stages of a plant community found in a particular area.
- The knowledge of ecological succession helps to understand the controlled growth of one or more species in a forest.
- Utilizing the knowledge of succession, even dams can be protected by preventing siltation.
- It gives information about the techniques to be used during reforestation and afforestation.
- It helps in the maintenance of pastures.
- Plant succession helps to maintain species diversity in an ecosystem.
- Patterns of diversity during succession are influenced by resource availability and disturbance by various factors.
- Primary succession involves the colonization of habitat of an area devoid of life.
- Secondary succession involves the reestablishment of a plant community in disturbed area or habitat.
- Forests and vegetation that we come across all over the world are the result of plant succession.

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## 7.4 Vegetation

Vegetation refers to the plant cover of an area. Geographically, India is a tropical country and also has strong monsoon climate and differs from other tropical regions of the World. India has four major climatic regions such as wet zone, intermediate zone, dry zone and arid zone, These regions are characterized by different types of natural vegetation. Nature of vegetation is also determined by several factors like altitude, types of plants, animals, climate, soil type, etc. Vegetation in Indian sub-continent is influenced by biotic factors and the existing human culture for a long time. The influence of man on plant formation and distribution is called anthropogenic effect on vegetation.

Tamil Nadu has a rich biodiversity right from the Gulf of Mannar to Western Ghats. Tamil Nadu shares the Western Ghats with states of Kerala, Karnataka, Goa, Maharashtra, Gujarat while Eastern Ghats is shared with the State of Andhra Pradesh. Of the 10 geographic zones in India, Coramandel (or) East Coast and Western Ghats are from Tamil Nadu.

### 7.4.1 Vegetation types of India and Tamil Nadu

Vegetation of India and Tamil Nadu consists of variety of plant communities and also possesses rich bio-diversity. It is classified in to the following four types, Which are explained with reference to their unique characteristics and distribution in India and Tamil Nadu:



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